

Session 1: Welcome to the IT World.

Task 1: 5 IT Job Roles and Descriptions

- **Software Developer:** Writes code to build, maintain, and upgrade software applications and systems.
- **QA Tester:** Evaluates software to find bugs, defects, and glitches before the product is released to users.
- **UI/UX Designer:** Designs the visual layout and user experience to ensure the software is intuitive and easy to use.
- **Data Analyst:** Inspects and models data to discover useful information and help businesses make smarter decisions.
- **AI Engineer:** Builds and trains artificial intelligence models and machine learning algorithms to automate complex tasks.

Task 2: Product vs. Service Companies

Product Companies (Create and sell their own software)

1. **Google:** Develops the Android operating system and Google Search engine used globally.
2. **Zoho:** Creates a suite of cloud-based business software tools like Zoho CRM for managing customer relationships.
3. **Microsoft:** Builds and licenses the Windows operating system and productivity tools like Microsoft Office.

Service Companies (Provide IT expertise and build software for other clients)

1. **TCS (Tata Consultancy Services):** Manages and upgrades the digital banking infrastructure for global banks.
2. **Infosys:** Develops custom e-commerce platforms and cloud migration systems for retail clients.
3. **Accenture:** Provides strategic IT consulting and manages cybersecurity operations for healthcare corporations.

Task 3: Full Forms and Uses

- **GitHub**
 - **Full Form:** GitHub (Proper noun, no official acronym).
 - **Use:** A web-based platform used by developers to store, track, and collaborate on software code using Git.
- **Slack**
 - **Full Form:** Search Log of All Conversation and Knowledge (originally, though now marketed as a proper noun).
 - **Use:** A channel-based messaging platform used by teams for instant communication and daily collaboration.
- **Jira**

- **Full Form:** Derived from the Japanese word for Godzilla (*Gojira*), used as a proper noun.
- **Use:** An issue and project tracking tool used by agile teams to plan, track, and manage software releases.
- **IDE**
 - **Full Form:** Integrated Development Environment.
 - **Use:** A software application (like VS Code) that provides a code editor, compiler, and debugger in one place for programmers.
- **API**
 - **Full Form:** Application Programming Interface.
 - **Use:** A set of rules that allows different software applications to communicate and share data with one another.

Task 4: 5 Programming Languages and Use Cases

Language	Popular Use Case
Python	Widely used for Data Science, Machine Learning, and AI development.
JavaScript	The primary language used to create interactive and dynamic elements on web pages.
Java	Heavily used for building large-scale Enterprise Applications and Android apps.
C++	Popular in Game Development and building high-performance system software.
Swift	The standard programming language used to build applications for Apple iOS and macOS.

Task 5: Difference Between a Developer and a Tester

The Difference:

A **Developer** is responsible for writing the code and building the software application from scratch based on requirements. Conversely, a **Tester** evaluates that completed code to identify bugs and ensure the software functions correctly before deployment.